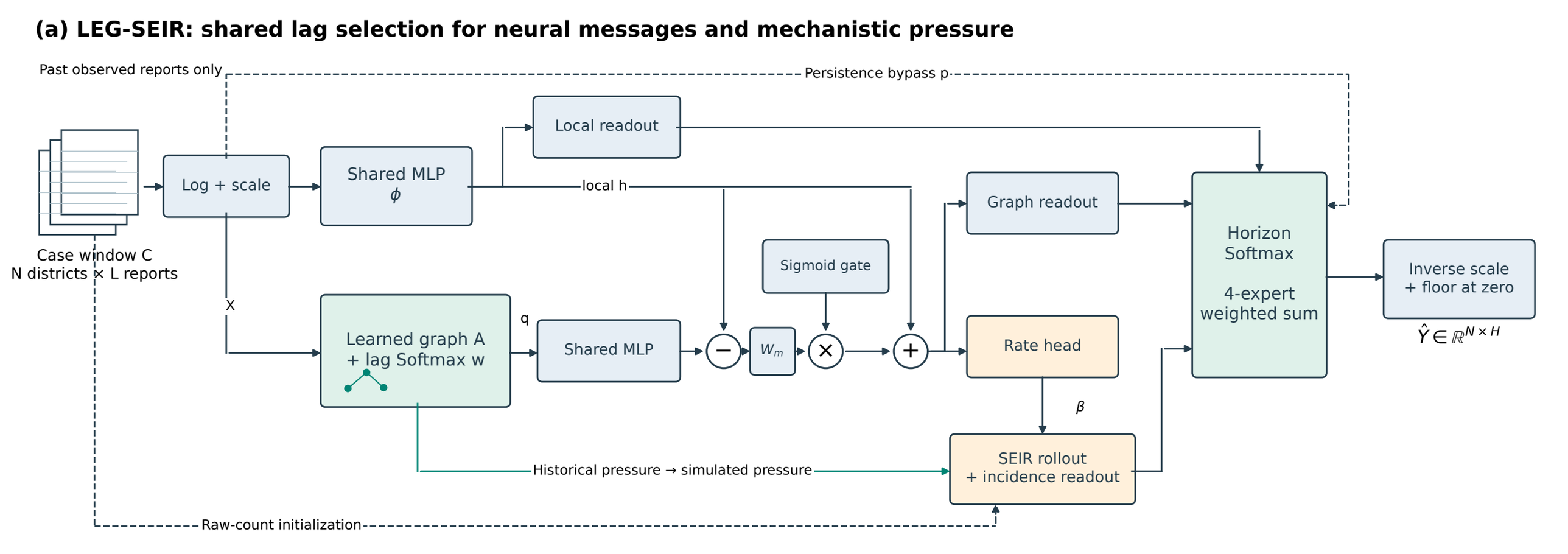
(a) LEG-SEIR: shared lag selection for neural messages and mechanistic pressure
Past observed reports only
Persistence bypass p
Case window C
N districts × L reports
Log + scale
Shared MLP
ϕ
Local readout
local h
Graph readout
Horizon
Softmax
4-expert
weighted sum
Inverse scale
+ floor at zero
$\hat{Y} \in \mathbb{R}^{N \times H}$
X
Learned graph A
+ lag Softmax w
q
Shared MLP
Sigmoid gate
w_m
Rate head
β
SEIR rollout
+ incidence readout
Historical pressure → simulated pressure
Raw-count initialization